

Lecture 21/Module 3

Sinusoidal Steady State Analysis

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Today's Topics

- Introduction to ac Circuits
- Sinusoids

Sinusoids

- A sinusoid is a signal that has the form of the sine or cosine function.
- A general expression for the sinusoid

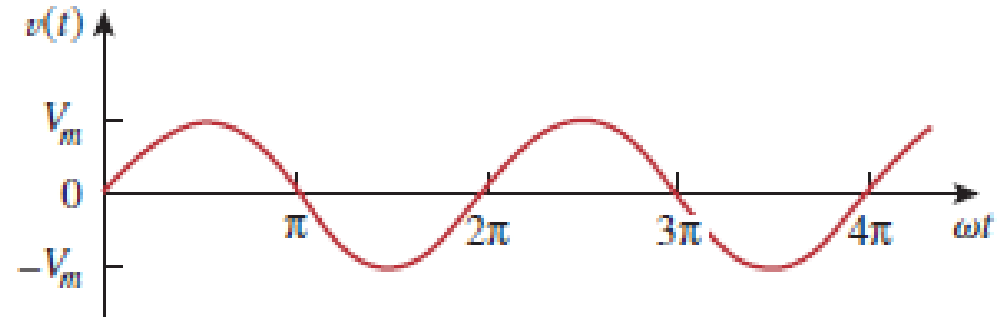
$$v(t) = V_m \sin \omega t$$

Where,

V_m = the amplitude of the sinusoid.

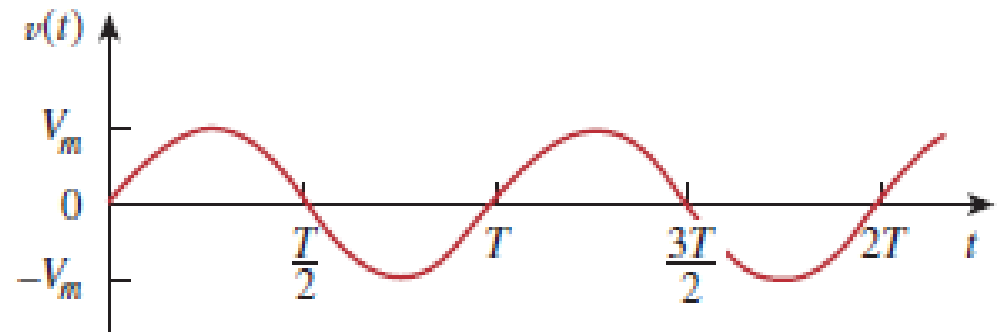
ω = the angular frequency in radians/s

ωt = the argument of the sinusoid



It is evident that the sinusoid repeats itself every T seconds; thus, **T is called the period of the sinusoid.**

we observe that $\omega T = 2\pi$.



$$T = \frac{2\pi}{\omega}$$

Sinusoids

The fact that $v(t)$ is repeating for every T seconds can be shown by replacing t by $t+T$.

$$\begin{aligned}v(t + T) &= V_m \sin \omega(t + T) = V_m \sin \omega \left(t + \frac{2\pi}{\omega} \right) \\&= V_m \sin(\omega t + 2\pi) = V_m \sin \omega t = v(t)\end{aligned}$$

$$v(t + T) = v(t)$$

This means $v(t)$ is periodic. In general, $v(t)$ has the same value at $v(t+T)$ and it is said to be periodic.

A periodic function is one that satisfies $f(t) = f(t+nT)$ for all t and for all integers n .

The period T of the periodic function is the time of one complete cycle or the number of seconds per cycle.

The reciprocal of this quantity is the number of cycles per second, known as the cyclic frequency f of the sinusoid.

$$f = \frac{1}{T}$$

$$\omega = 2\pi f$$

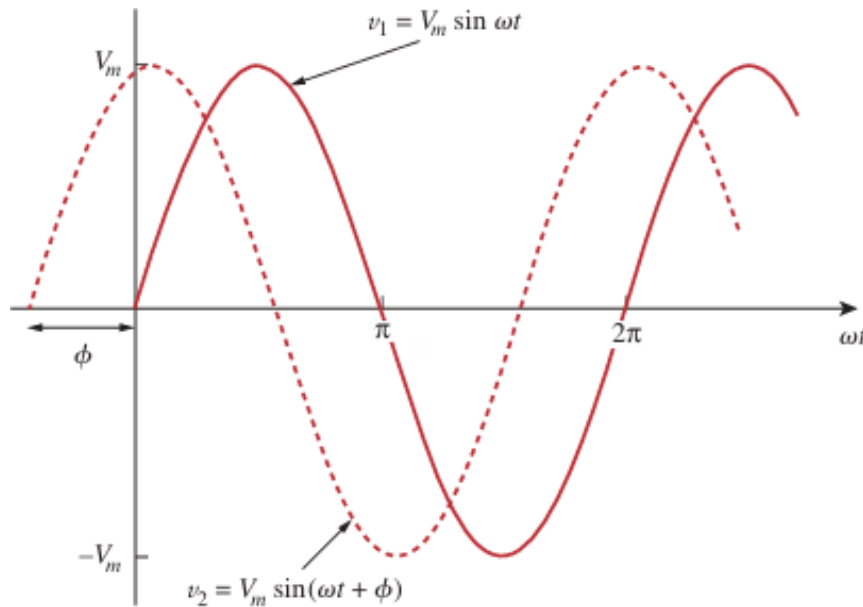
ω is in radians per second (rad/s), f is in hertz (Hz).

Sinusoids

Let us now consider a more general expression for the sinusoid.

$$v(t) = V_m \sin(\omega t + \phi)$$

Where ωt is the argument and ϕ is the phase. Both argument and phase can be in radians or degrees. Consider, $v_1(t) = V_m \sin \omega t$ and $v_2(t) = V_m \sin(\omega t + \phi)$



The starting point of v_2 occurs first in time. Therefore, we say that v_2 leads v_1 by ϕ or that v_1 lags v_2 by ϕ .

If we also say that v_1 and v_2 are out of phase. If $\phi = 0$, then v_1 and v_2 are said to be in phase; they reach their minima and maxima at exactly the same time, which means they operate at the same frequency;

They do not need to have the same amplitude.

A sinusoid can be expressed in either sine or cosine form.

Sinusoids

Trigonometric identities: Using these relationships, we can transform a sinusoid from sine form to cosine form or vice versa.

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

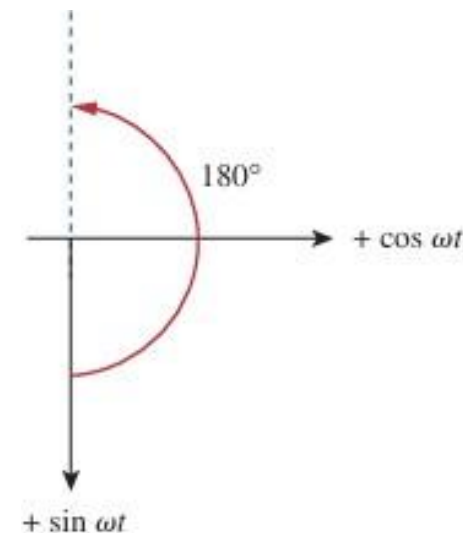
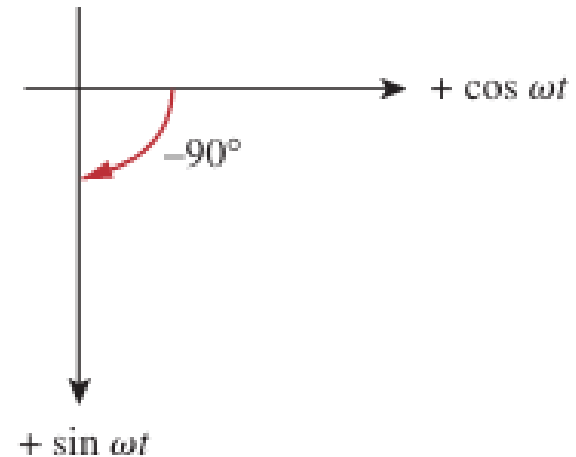
$$\sin(\omega t \pm 180^\circ) = -\sin \omega t$$

$$\cos(\omega t \pm 180^\circ) = -\cos \omega t$$

$$\sin(\omega t \pm 90^\circ) = \pm \cos \omega t$$

$$\cos(\omega t \pm 90^\circ) = \mp \sin \omega t$$

Subtracting 90° from the argument of $\cos \omega t$ gives $\sin \omega t$.



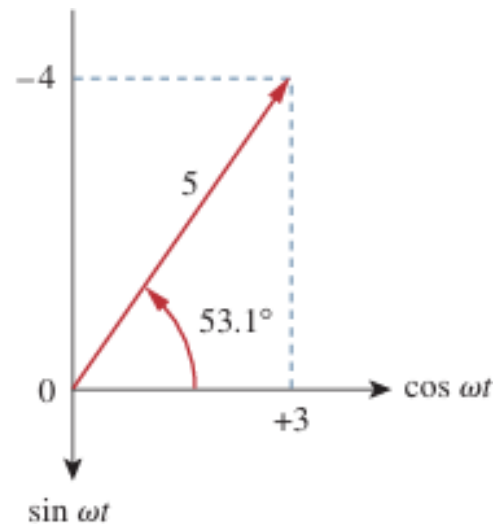
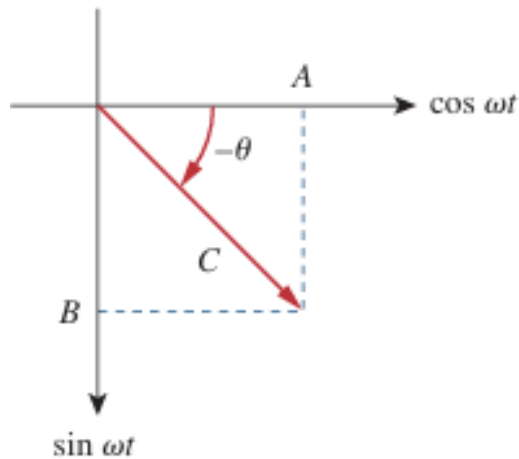
Sinusoids

Adding 2 sinusoid $A \cos \omega t + B \sin \omega t = C \cos(\omega t - \theta)$

$$C = \sqrt{A^2 + B^2}, \quad \theta = \tan^{-1} \frac{B}{A}$$

For example,

$$3 \cos \omega t - 4 \sin \omega t = 5 \cos(\omega t + 53.1^\circ)$$



Sinusoids

1. Find the amplitude, phase, period, and frequency of the sinusoid $v(t) = 12 \cos(50t + 10^\circ)$

The amplitude = 12 V

The phase = 10°

The angular frequency = 50 rad/s

The period = 0.1257 s

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{50}$$

The frequency is $f = 1/T = 7.958$ Hz.

2. Given the sinusoid $30 \sin(4\pi t - 75^\circ)$, calculate its amplitude, phase, angular frequency, period, and frequency.

Ans: 30, 12.57 rad/s, 0.5 s, 2 Hz

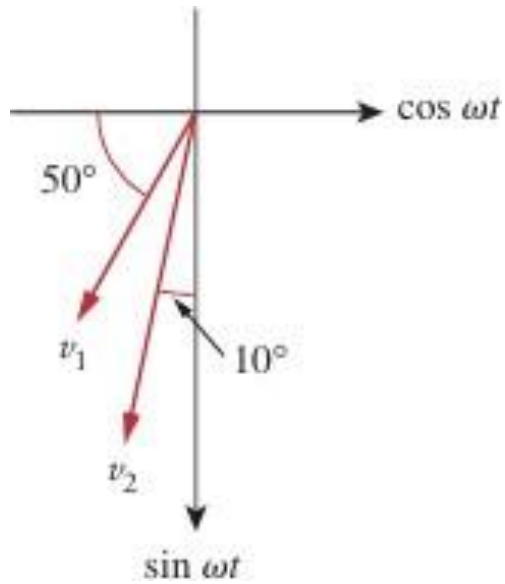
3. Calculate the phase angle between $v_1 = -10 \cos(\omega t + 50^\circ)$ and $v_2 = 12 \sin(\omega t - 10^\circ)$. State which sinusoid is leading.

$$v_1 = -10 \cos(\omega t + 50^\circ) = 10 \cos(\omega t + 50^\circ - 180^\circ)$$

$$v_1 = 10 \cos(\omega t - 130^\circ) \text{ or } v_1 = 10 \cos(\omega t + 230^\circ)$$

$$v_2 = 12 \sin(\omega t - 10^\circ) = 12 \cos(\omega t - 10^\circ + 90^\circ)$$

Sinusoids



v_2 leads v_1 by 30° .

Thank You